



Related Rates: Advanced Applications

Calculus Worksheet · Grade 11-12

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Learning Objectives

- Apply implicit differentiation with respect to time to geometric formulas
- Set up related rates problems using the Pythagorean theorem and other geometric relationships
- Solve for unknown rates of change by substituting given values

For each problem, draw a diagram, identify the given rate and the rate to find, write the equation, differentiate implicitly with respect to time, and solve.

1. A 10 ft ladder rests against a vertical wall. The bottom slides away from the wall at 1 ft/s. How fast is the top sliding down when the bottom is 6 ft from the wall?

$$x^2 + y^2 = 10^2, \quad \frac{dx}{dt} = 1$$

Answer: _____

2. A 13 ft ladder leans against a wall. The base slides away from the wall at 2 ft/s. How fast is the top descending when the base is 5 ft from the wall?

$$x^2 + y^2 = 13^2, \quad \frac{dx}{dt} = 2$$

Answer: _____

3. Air is pumped into a spherical balloon so that its volume increases at 100 cubic cm per second. How fast is the radius increasing when the radius is 25 cm?

$$V = \frac{4}{3}\pi r^3, \quad \frac{dV}{dt} = 100$$

Answer: _____

4. A circular oil slick expands so its radius grows at 0.5 m/s. How fast is the area increasing when the radius is 8 m?

$$A = \pi r^2, \quad \frac{dr}{dt} = 0.5$$

Answer: _____

5. A cube's edges expand at 3 cm/s. How fast is the volume increasing when the edge length is 10 cm?

$$V = s^3, \quad \frac{ds}{dt} = 3$$

Answer: _____



6. A 15 ft ladder slides down a wall. The top descends at 2 ft/s. How fast is the bottom moving away when the top is 9 ft above the ground?

$$x^2 + y^2 = 15^2, \quad \frac{dy}{dt} = -2$$

Answer: _____

7. Water is poured into a conical tank (radius 4 m, height 8 m) at 2 cubic m per minute. How fast is the water level rising when the depth is 3 m? Use the relation $r = h/2$.

$$V = \frac{1}{3}\pi r^2 h = \frac{\pi h^3}{12}$$

Answer: _____

8. Two cars leave the same point. Car A travels north at 60 mph and Car B travels east at 80 mph. How fast is the distance between them changing after 1 hour?

$$z^2 = x^2 + y^2, \quad \frac{dx}{dt} = 80, \quad \frac{dy}{dt} = 60$$

Answer: _____

9. Sand falls into a conical pile so its height equals its base radius. The volume increases at 10 cubic ft per minute. How fast is the height rising when the height is 5 ft?

$$V = \frac{1}{3}\pi r^2 h = \frac{\pi h^3}{3}$$

Answer: _____





Emphasize that students should differentiate before substituting numerical values, and remind them to memorize common geometric formulas like area of a circle and volume of a sphere.

Solutions

1. A 10 ft ladder rests against a vertical wall. The bottom slides away from the wall at 1 ft/s. How fast is the top sliding down when the bottom is 6 ft from the wall?

$$x^2 + y^2 = 10^2, \quad \frac{dx}{dt} = 1$$

- Draw a right triangle with the wall as the vertical leg and the ground as the horizontal leg.
- Use the Pythagorean theorem: x squared plus y squared equals 100.
- Differentiate both sides with respect to time to get $2x$ times dx/dt plus $2y$ times dy/dt equals zero.
- When x equals 6, find y equals 8 from the Pythagorean theorem.
- Substitute x equals 6, y equals 8, and dx/dt equals 1, then solve for dy/dt to get negative three-fourths ft per second.

Answer: $\frac{dy}{dt} = -\frac{3}{4}$ ft/s

2. A 13 ft ladder leans against a wall. The base slides away from the wall at 2 ft/s. How fast is the top descending when the base is 5 ft from the wall?

$$x^2 + y^2 = 13^2, \quad \frac{dx}{dt} = 2$$

- Set up the Pythagorean relationship with hypotenuse 13.
- Differentiate implicitly with respect to time.
- When x equals 5, solve for y to get 12.
- Substitute the known values into the differentiated equation.
- Solve for dy/dt to obtain negative five-sixths ft per second.

Answer: $\frac{dy}{dt} = -\frac{5}{6}$ ft/s

3. Air is pumped into a spherical balloon so that its volume increases at 100 cubic cm per second. How fast is the radius increasing when the radius is 25 cm?

$$V = \frac{4}{3}\pi r^3, \quad \frac{dV}{dt} = 100$$

- Write the volume formula for a sphere.
- Differentiate both sides with respect to time to get dV/dt equals $4\pi r$ squared times dr/dt .
- Substitute dV/dt equals 100 and r equals 25.
- Solve for dr/dt to get 1 over 25π cm per second.

Answer: $\frac{dr}{dt} = \frac{1}{25\pi}$ cm/s



4. A circular oil slick expands so its radius grows at 0.5 m/s. How fast is the area increasing when the radius is 8 m?

$$A = \pi r^2, \quad \frac{dr}{dt} = 0.5$$

→ Use the area formula for a circle.

→ Differentiate with respect to time to get dA/dt equals 2 pi r times dr/dt .

→ Substitute r equals 8 and dr/dt equals one-half.

→ Compute dA/dt equals 8 pi square meters per second.

Answer: $\frac{dA}{dt} = 8\pi \text{ m}^2/\text{s}$

5. A cube's edges expand at 3 cm/s. How fast is the volume increasing when the edge length is 10 cm?

$$V = s^3, \quad \frac{ds}{dt} = 3$$

→ Write the cube volume formula V equals s cubed.

→ Differentiate to get dV/dt equals 3 s squared times ds/dt .

→ Substitute s equals 10 and ds/dt equals 3.

→ Compute dV/dt equals 900 cubic cm per second.

Answer: $\frac{dV}{dt} = 900 \text{ cm}^3/\text{s}$

6. A 15 ft ladder slides down a wall. The top descends at 2 ft/s. How fast is the bottom moving away when the top is 9 ft above the ground?

$$x^2 + y^2 = 15^2, \quad \frac{dy}{dt} = -2$$

→ Apply the Pythagorean theorem with hypotenuse 15.

→ Differentiate implicitly to get $2x \, dx/dt$ plus $2y \, dy/dt$ equals zero.

→ When y equals 9, solve for x to get 12.

→ Substitute the values and solve for dx/dt to obtain three-halves ft per second.

Answer: $\frac{dx}{dt} = \frac{3}{2} \text{ ft/s}$

7. Water is poured into a conical tank (radius 4 m, height 8 m) at 2 cubic m per minute. How fast is the water level rising when the depth is 3 m? Use the relation $r = h/2$.

$$V = \frac{1}{3}\pi r^2 h = \frac{\pi h^3}{12}$$

→ Express r in terms of h using similar triangles: r equals h over 2.

→ Substitute into the cone volume formula to get V equals pi h cubed over 12.

→ Differentiate with respect to time to get dV/dt equals pi h squared over 4 times dh/dt .

→ Substitute dV/dt equals 2 and h equals 3, then solve for dh/dt to get 8 over 9 pi m per min.

Answer: $\frac{dh}{dt} = \frac{8}{9\pi} \text{ m/min}$



8. Two cars leave the same point. Car A travels north at 60 mph and Car B travels east at 80 mph. How fast is the distance between them changing after 1 hour?

$$z^2 = x^2 + y^2, \quad \frac{dx}{dt} = 80, \quad \frac{dy}{dt} = 60$$

→ Set up the Pythagorean relationship for the distance z between the cars.

→ Differentiate implicitly to get $2z \, dz/dt$ equals $2x \, dx/dt$ plus $2y \, dy/dt$.

→ After 1 hour, x equals 80, y equals 60, so z equals 100.

→ Substitute the values and solve for dz/dt to get 100 mph.

Answer: $\frac{dz}{dt} = 100$ mph

9. Sand falls into a conical pile so its height equals its base radius. The volume increases at 10 cubic ft per minute. How fast is the height rising when the height is 5 ft?

$$V = \frac{1}{3}\pi r^2 h = \frac{\pi h^3}{3}$$

→ Since r equals h , substitute into the cone volume formula to get V equals πh^3 over 3.

→ Differentiate with respect to time to get dV/dt equals πh^2 times dh/dt .

→ Substitute dV/dt equals 10 and h equals 5.

→ Solve for dh/dt to obtain 2 over 5 π ft per minute.

Answer: $\frac{dh}{dt} = \frac{2}{5\pi}$ ft/min

